# Thank you

for celebrating our student researchers.

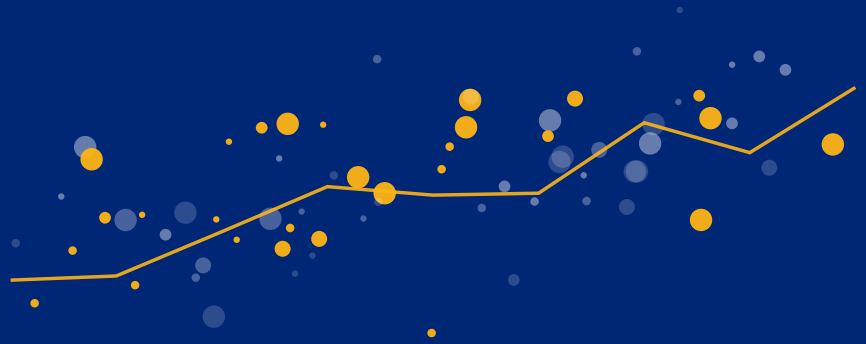
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UC Berkeley College of Computing, Data Science, and Society

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CDSS DATA DISCOVERY

# Symposium

*Fall 2025*

**Wednesday, December 10, 2025**

1:30 p.m. – 3:00 p.m.

Lobby of the Hearst Memorial Mining Building

UC Berkeley

PROGRAM

cdss.berkeley.edu/discovery

FALL 2025 SYMPOSIUM

# Welcome

The Data Discovery Program connects teams of Berkeley students with mentors across campus and beyond, using data to understand real-world problems in academia, industry, government and nonprofits.

The Symposium is the culmination of the semester as student researchers share their work through poster presentations. Please join us in celebrating their contributions to the data research community.

You can visit our webpage at https://cdss.berkeley.edu/discovery or email us at cdss-datadiscovery@berkeley.edu.

SCHEDULE

## Wednesday, December 10, 2025

Lobby of the Hearst Memorial Mining Building, UC Berkeley

**1:30 p.m.** — **Poster session begins**

**2:00 p.m.** — **Keynote address**
Associate Dean of Students, Professor Narges Norouzi

**2:15 p.m.** — **Presentation of student awards**

**3:00 p.m.** — **Event concludes**

AT A GLANCE

**48**
RESEARCH POSTERS

**6**
AWARD RECIPIENTS



*Notes*

## *Notes*



CELEBRATING EXCELLENCE

# Award Recipients

AWARDED FOR RESEARCH STORYTELLING

### Automated Oxford Nanopore DNA Sequencing Pipeline

Jaina Gandhi, Bashar Kabbarah, Reina Kushihashi, Jennifer Sun and Alana Waterman

*Mentored by Scott Geller, Brian McCarthy, UC Berkeley DNA Sequencing Facility*

AWARDED FOR RESEARCH STORYTELLING

### Coping with AI: Faculty Sentiments and AI Policies Across Course Syllabi

Christopher Mach, Lynn Chien, Smrithi Senthilnathan and Sophie McKenna

*Mentored by Igor Chirikov, Berkeley Center for Studies in Higher Education*

AWARDED FOR METHODS: STATISTICAL INFERENCE

### Continuous Glucose Monitors for Gestational Diabetes

Allen M. Zhou and Nathaniel E. Santoso

*Mentored by Dr. Michael A. Kohn and Dr. Nasim C. Sobhani, UC San Francisco*

AWARDED FOR METHODS: LARGE LANGUAGE MODELING

### Quantifying Market Sentiment: A Multi-Model Analysis of Wall Street Reports

Amy Tran, Holden Carrillo, Martin Li and Amber Yu

*Mentored by Ethan Yen, Gilead Sciences*

HONORABLE MENTION FOR POSTER DESIGN

### Energy Efficiency and Reliability Exploration: Ride Electric Bus Fleet in Anaheim

Brian Whitney and Jacqueline Vo

*Mentored by Timothy Lipman and Matthew Elke, Transportation Sustainability Research Center*

HONORABLE MENTION FOR POSTER DESIGN

### Safety Evaluation of Autonomous Vehicles in Real-World Settings

Scarlet Shi and Neel Panchumarthi

*Mentored by Peggy Wang, California PATH Research Center at UC Berkeley Institute for Transportation Studies*

48 RESEARCH PROJECTS

# Poster Session

Student research teams present their work alongside the mentors and organizations who made it possible.

### Health Community Data Optimization & Engagement Platform
Maryum Syed, Matthew Ho, Collin Nguyen, Joshua Davis, Nia Shelton
*Mentored by Jolie Lam, CITRIS and the Banatao Institute - UC Berkeley*

### Analyzing AI Policies Across College Syllabi
Christopher Mach, Lynn Chien, Smrithi Senthilnathan, Sophie McKenna
*Mentored by Igor Chirikov, Berkeley Center for Studies in Higher Education*

### Automated Deep-Learning Segmentation and Density Estimation of Polymetallic Nodules in Deep-Sea Seafloor Imagery
Manasvi Lodha, Inez Guadalupe Alvarez, Hitha Sri Varganti, Vihaan Khandelwal
*Mentored by Kimberly Baldwin, Bureau of Ocean Energy Management*

### Using Reinforcement Learning to Reduce Energy Waste for the Drying Process of a Wood Panel Manufacturer
Emily Hsu, Abraham Guan, Rohan Bhagavan
*Mentored by Ricardo Perez, Umbo Energy*

### FLS Labeling - Hard Negative Mining for Sentence Transformers
Aditi Bordia, Tiana Zhou, Yi Hong Ng
*Mentored by Yu Shen, BART*

### Continuous Glucose Monitors for Gestational Diabetes
Nathaniel Santoso, Allen Zhou
*Mentored by Dr. Michael Kohn, Dr. Nasim Sobhani, University of California, San Francisco (UCSF)*

### Analyzing California K12 Facility Master Plans With LLMs
Sierra Meisel, Hamza Khalilullah
*Mentored by Sara Hinkley, Center for Cities + Schools*

### Knowledge graph question answering for FDA 510(k) submissions
Collin Lee, William Wu, Evelyn Shen
*Mentored by Joey Yip, Medhive Ai*

### Tracking Inequities in California School Facility Funding
Kristen Li, Avery Schmit-Hipolito
*Mentored by Sara Hinkley, Center for Cities + Schools*



WITH GRATITUDE

# Acknowledgments

We would like to thank everyone who's contributed to the College of Computing, Data Science, and Society's Data Discovery program and making the Symposium come to life. Data Discovery would not be possible without the leadership of Professor Deb Nolan. Sierra J. White-Koutsoubos and Associate Dean of Students Professor Narges Norouzi have provided invaluable guidance. Jessica McDaniels arranged crucial details of the event, and Camille Simsuangco Roxas helped execute it. CDSS Development contributed funds for the Symposium. Claire Chen, Keryssa Li and Leo Liao have helped keep Data Discovery running smoothly this semester. Ryan Lovett helped arrange access to computing resources and provided technical support, and we acknowledge the National Science Foundation's National Artificial Intelligence Research Resource Pilot for providing computational resources (award number NAIRR240387). Thank you to the experts who shared their time and expertise through guest lectures and panels for the Data Discovery course. Finally, we'd like to thank the Data Discovery mentors who provide the projects and mentorship that make the program possible.

### IRIS
Hanjo Kim, Anupriya Islam, Gilbert Harijanto, Andre Ramos
*Mentored by Ryan Lingo, Honda Research Institute USA*

### Redesigning an entity-linking model
Tanish Kher, Purva Marfatia, Nathan Tran
*Mentored by Adam Anderson, Collin Duong, UC Berkeley D-Lab*

### Human vs. LLM Text Simplification for K-12 Learners
Khue N Do, Holden Fees, Senula James Wijeratne, Siya Patel
*Mentored by Anastasia V. Smirnova, San Francisco State University - ECOLE Lab*

### Automating Anti-Displacement Policy Discovery: An LLM-Powered Web Scraping Solution
Allen Lopez Iyer, Ariana Siordia, Bo Wang, Zack Yu
*Mentored by Karen Chapple, Madeleine Parker, Chenghao Li, Urban Displacement Center*

### AI Peer Reviewer Matching
Kovidh Singh
*Mentored by Hildy Baker, RR\ID*

### Impacts of Operational Decisions and a Changing Climate on California's Water Future
Isabelle Wang, Rain Zou, Isabelle Goebel, Sabi Can Ruso
*Mentored by Dino Bellugi, James Gilbert, Laurel Larsen, UC Berkeley Department of Geography*



### FDA Citizen Petition Analysis
Adithya Ayanam, Herman Ho, Kyaw Min Khant, Smriti Rangarajan
*Mentored by Allison Schmitt, University of Oregon School of Law*

### Wildfire Data Repository for the East Bay Hills
Nicolas Bonville, Elise Chang, Sosie Casteel, Nicolas Sapountzis, Andrew Shoenen
*Mentored by Alec Dara-Abrams, Berkeley FireSafe*

### Interpreting AlphaEarth's Embedding Space for Environmental Monitoring
Aayan Rizvi, Eric Yu
*Mentored by Huiqi Wang, The Keenan Group*

### Drug Discovery Lead Optimization with Agents and Deep Learning
Alivia Ding, Caroline Chen, Katherine Ge, Rhythm Goyal, Viraj Roy Gupta
*Mentored by Alan Cheng, Alec Glisman, David Li He, Xiaodong Zhu, Merck & Co*

### Coursewise AI Support Chatbot
Miguel Contreras, Ethan Zhang, Michelle Li, Arnav Agarwal
*Mentored by Angikaar Chana, Equivalence Systems*

### Mineralog: Mineral Composition Analysis for Geological Characterization
Kevin Geng, Raghavan Srinivas, Rebecca Bachtra
*Mentored by Kimberly Baldwin, Kevin Smith, Bureau of Ocean Energy Management*

### Carbon Sustain: Sustainability Awareness and Compliance through Tools and Data
Tiffany Yeh, Anica Chen, Daniel Li, Cedric Xu
*Mentored by Paul Bryzek, CarbonSustain.io*

### Trip Mode Segmentation, Classification, and Forecasting for Verified Carbon Credits
William Chang, Shiama Srikantan, Anthony Gomez-Mier, Dylan Stier
*Mentored by Aurora Khatibi Garrity, HERO2*

### Hallucinations Detection in Medical Device Question Answering
Mark Shi, Amber Tien, Sean Yue, Arhaan Aggarwal
*Mentored by Joey Yip, MedHive*

### Identifying Anomaly Aurora with Computer Vision Models
Tommy Duong, Alex Toohey, Sabrina Nazarzai
*Mentored by Brian Harding, Claire Gasque, UC Berkeley Space Sciences Laboratory*

### Safety Analysis of Autonomous Vehicles in Real-World Settings
Neel Panchumarthi, Scarlet Shi
*Mentored by Dr. Peggy Wang, UC Berkeley Institute for Transportation Studies - California PATH Research Center*

### Quantifying carbon dioxide emissions from U.S. power plants using deep learning and simulated satellite data

Deleena Ghosh, Liberty Lauer, Rohin Juneja

*Mentored by Prof. Ronald Cohen, Hikari Murayama, Naomi Asimow, Cohen Research Group*

### Interactive visual dashboard for Microbial Communities

Haaziq Kazi, Lia Shamir, Rohan Panchal, Fion Wu, Annie Gao

*Mentored by Jennifer Kuehl, Arkin Lab*

### Geopogo AI

Sanatan Mishra, Akshay Sunkara, Chenghao Li, Yuanbo Pang

*Mentored by Mike Hoppe, Dave Alpert, Geopogo*

### Predicting Student Chronic Absenteeism

Joshua Xiong, Erin Pan, Angel Tovar, Alex Zhai

*Mentored by Ana Costa, Oakland Natives Give Back*

### CourseWise – Competitive Analysis and Candidate Metrics

Emma Cao, Amelia Jin, Geoffrey Kim, John Hu

*Mentored by Angikaar Chana, CouseWise*

### Improving VerifAI Decision Quality

Keyue Liao, Victoria Fengyi Ruan, Ellie Wang, Amanda Wu, Zoe Yip

*Mentored by Nikolaus Wischmann, GrapeData*

### C. Light Retina Agent

Diana Tao, Jenny Shang, Yash Thapliyal, Yuki Chen

*Mentored by Yifan Zhang, C. Light technology*

### Expanding Our Understanding of Environmental and Climate Action in Schools; Emphasis on Future Goals and Funding

Aniruthan Ravisankar, Jesus Ochoa, Ojas Sathaye, Perseus Zhao

*Mentored by Devin Ngo, Sarah Whiting, Judy He, Ten Strands*

### Enhancing RR\ID Editorial Workflows: Improving Reviewer Quality-Check Portal and Conflict-of-Interest Detection Tool

Riley Roggenkamp, Tracy Lian, Dominic Yap

*Mentored by Reily Fairchild, Viraj Bhosale, Rapid Reviews\Infectious Diseases (RR\ID)*

### Exploring Electric Bus Effectiveness in Anaheim, CA

Brian Whitney, Jacqueline Vo

*Mentored by Timothy Lipman, Matthew Elke, Transportation Sustainability Research Center (TSRC)*

### AI Governance Training for South Korean DPOs

Andre Leung, Deeksha Vaidyanathan, Erin Kim, Renta Oshima

*Mentored by Christine Fostvedt-Mills, HewardMills*



### Automating Oxford Nanopore DNA Sequencing Pipeline

Jaina Gandhi, Bashar Kabbarah, Reina Kushihashi, Jennifer Sun, Alana Waterman

*Mentored by Scott Geller, Brian McCarthy, UC Berkeley DNA Sequencing Facility*

### Why the FDA Takes So Long to Say No: A Computational Approach to Governmental Accountability

Abtin Khashayar, Anh Tran, Avani Kulshreshtha, Ashley Huang

*Mentored by Allison Schmitt, University of Oregon School of Law, UC Berkeley Center for Law & Technology*

### Sharpen DTX AI Powered Recommendation Engine: A Framework to Personalize Mental Health Support for At-Risk Children

Krish Bala, Pranay Gonuguntla, Pedro Ochoa, Dhruv Syngol, Claudia Wu

*Mentored by Tim Farrell, Robyn Hussen Farrel, Resiliency Technologies Inc.*

### Explainable Data Analytics: A Neurosymbolic AI Approach

Jason Park, Kaden Tang, Wayland La, Eric Li, Yinghan Sun, Daniel Firoozabadi

*Mentored by Naveen Ashish, Inferlink Corporation*

### Does Human Capital Matter?: Analysis Of The Impact Of Dedicated Staff On Environmental And Climate Action In Schools

Pratham Rangwala, Huy Nguyen, Uichan Lee

*Mentored by Judy He, Sarah Whiting, Ten Strands*

### CWN EmpowerHer

Jennifer Ren, Sarah Wan, Cindy Wang, Rebecca Kang

*Mentored by Jolie Lam, CITRIS and Cal Women's Network*

### Where are California's Green Career Pathways?

Adrian Mar

*Mentored by Sarah Whiting, Ten Strands*

### Quantifying Market Sentiment: A Multi-Model Analysis of Wall Street Reports

Amy Tran, Martin Li, Holden Carrillo, Amber Yu

*Mentored by Ethan Yen, Gilead Sciences*

### Improving BART Chatbot Using MemGPT Tool Calling

Ethan Qiu, Neil Kamath, Audrey Wong, Ray Lin

*Mentored by Yu Shen, Bay Area Rapid Transit (BART)*

### Exploring Time-Series Forecasting for Network Utilization in Scientific Data Flows

Hetvi Patel, Mehul Jaiswal, Hugo Fang

*Mentored by John Wu, Alex Sim, Lawrence Berkeley National Laboratory*

### Generative AI in Surface Water Anomaly Testing

Akansha Jain, Diego Gutierrez, Himani Balaga, Tony Getsin

*Mentored by Dan Wang, California Water Boards*